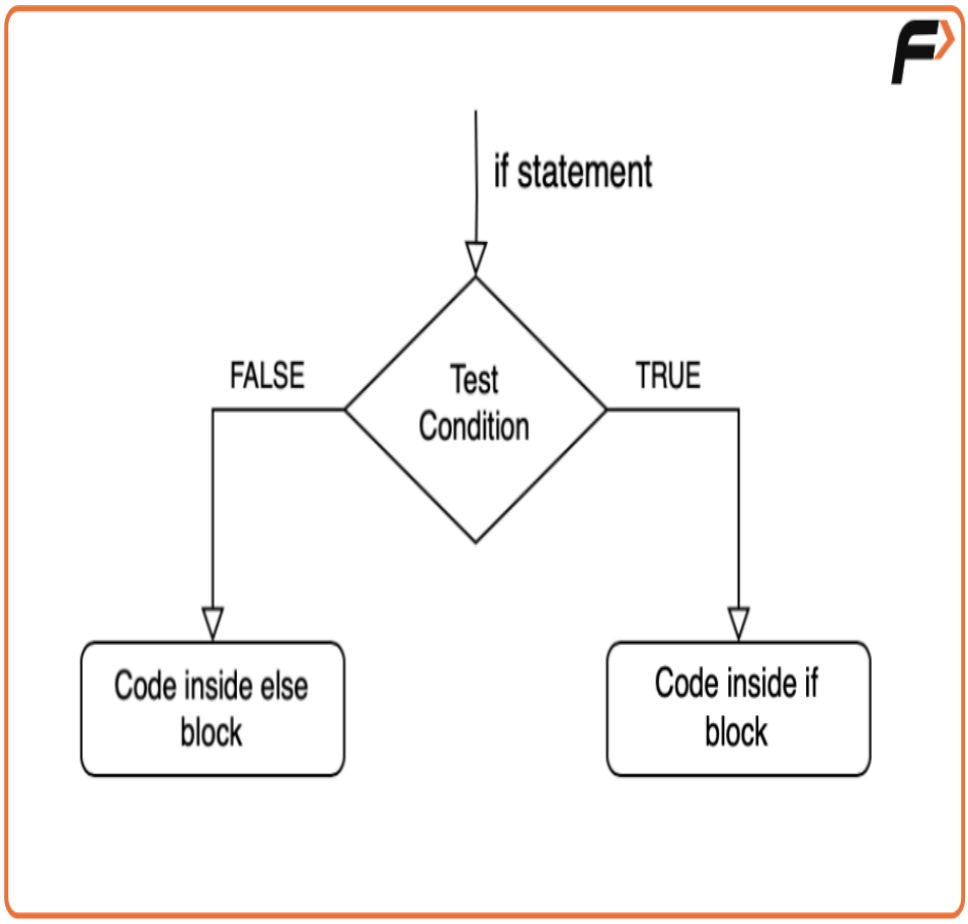
Conditional statements are a fundamental concept in programming that allows you to make decisions based on certain conditions. These statements enable your code to **execute different blocks of cod**e depending on whether **specific conditions** are met or not. In this blog post, we'll delve into the basics of **conditional statements**, starting with the ubiquitous **if-else** statement and gradually exploring more complex scenarios.

The `if-else` Statement



'if statement' is used to execute a block of code only if a certain condition is met. It allows us to conditionally execute code based on whether the specified condition is true.

'else statement', on the other hand, is an optional companion to the if statement. It specifies what code to execute if the condition in the if statement is not met (i.e. if it is false).

Let's break down the flow of control:

- If the test condition in the if statement is true, a block of code inside the if block will be executed.
- If the test condition is false, the code inside the else block (if present) will be executed.

Code:

C++JavaPythonJavaScript

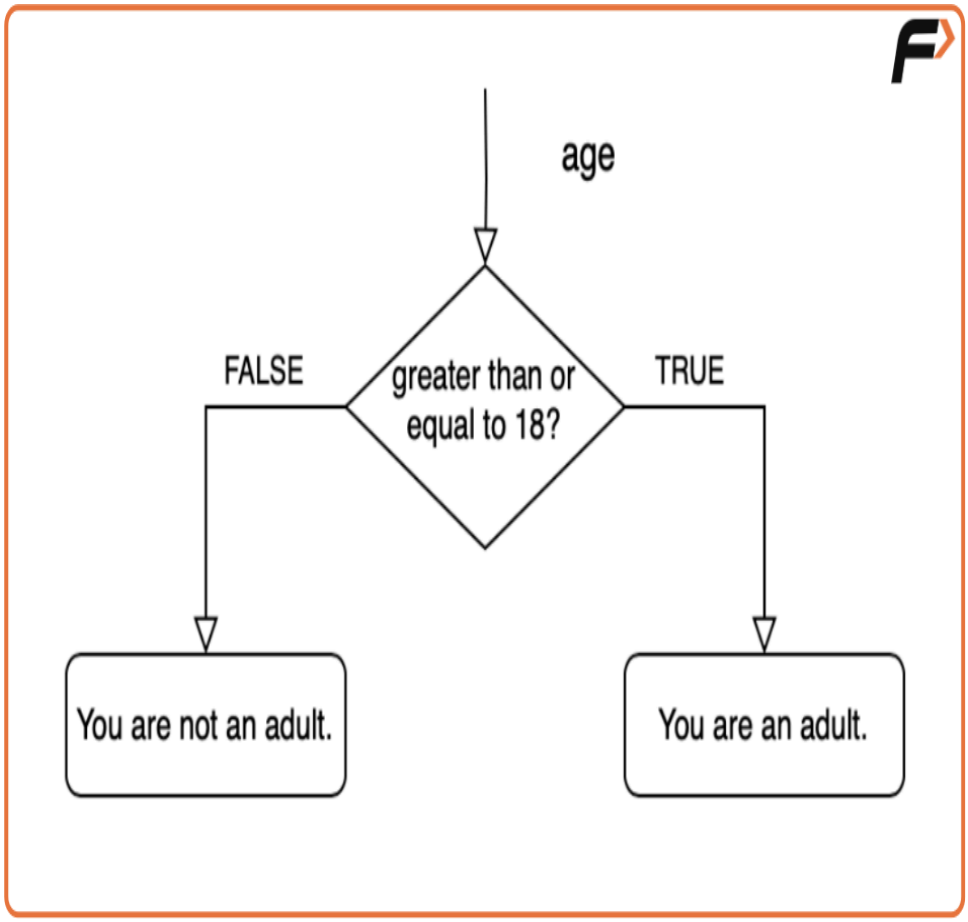
```
#include<bits/stdc++.h>
using namespace std;
int main() {
    // Declare a variable to store the user's age
    int age;

    // Ask the user to enter their age
    cout << "Enter your age: ";
    cin >> age;  // Take input from the user

    // Check if the user is 18 or older
    if (age >= 18) {
        // If true, print that the user is an adult
        cout << "You are an adult." << endl;
    } else {
```

```
        // Otherwise, print that the user is
not an adult
        cout << "You are not an adult." <<
endl;
    }

    return 0; // End of program
}
```



In this example, we take the **user's age as input** and use an **if statement** to check if the age is greater than or equal to 18. If the condition is true, it prints "You are an adult," and if it's false ie. the **'else'** condition is satisfied, it prints "You are not an adult."

Simplifying Code with "else if"

As your code becomes more complex, you'll often encounter scenarios where you need to check multiple conditions. Instead of writing multiple independent if statements, you can streamline your code using else if statements.

Let's say we want to grade students based on their marks within specific ranges:

Code:

C++JavaPythonJavaScript

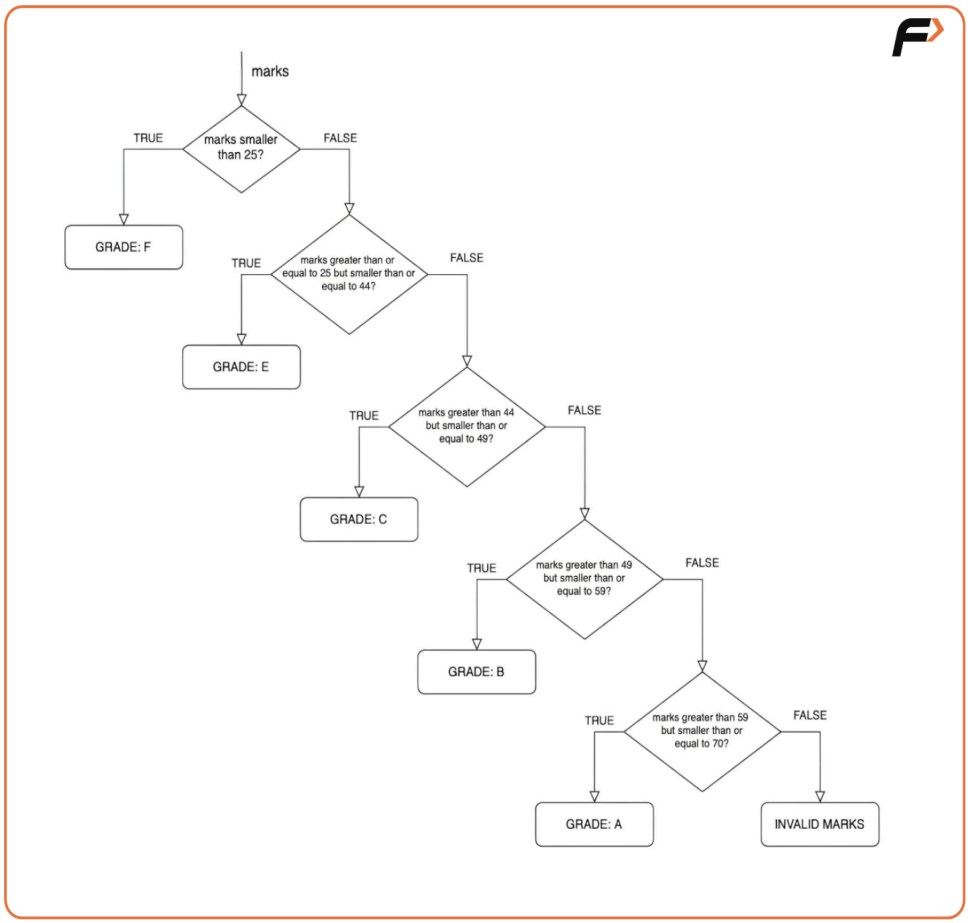
```
#include <iostream>
using namespace std;

int main() {
    // Declare and initialize marks variable
    int marks = 54;

    // Check grade conditions using if-else
ladder
    if (marks < 25) {
        cout << "Grade: F" << endl;  // Less
than 25 is Grade F
    } else if (marks >= 25 && marks <= 44) {
```

```
        cout << "Grade: E" << endl;  //
Between 25 and 44 is Grade E
    } else if (marks >= 45 && marks <= 49) {
        cout << "Grade: D" << endl;  //
Between 45 and 49 is Grade D
    } else if (marks >= 50 && marks <= 59) {
        cout << "Grade: C" << endl;  //
Between 50 and 59 is Grade C
    } else if (marks >= 60 && marks <= 69) {
        cout << "Grade: B" << endl;  //
Between 60 and 69 is Grade B
    } else if (marks >= 70) {
        cout << "Grade: A" << endl;  // 70
and above is Grade A
    } else {
        cout << "Invalid marks entered." <<
endl;  // Handles unexpected cases
    }

    return 0;
}
```



The provided code for grading based on marks is functional, but it **can be simplified** for better readability and maintainability. In the current code, there are **several redundant comparisons** of marks with specific values. For example, when checking for grades E, C, B, and A, you have to **repeatedly check** marks >= X && marks <= Y, which can be error-prone and hard to maintain as the grade ranges change.

We can refactor the code, we can **remove the lower bounds** and check only the **upper bounds** for each grade as the code flow is such that we move along the conditions only after satisfying the previous one.